

TANUSHREE NEPAL

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Research Interests: Retrieval-Augmented Generation · Biomedical NLP · LLM Reliability & Faithfulness · Information Retrieval · Recommender Systems · AI Safety & Fairness

EDUCATION

M.S. Data Science & Artificial Intelligence	University of Central Missouri, Lee's Summit, MO	Expected Dec 2026	GPA: 3.87/4.00
M.B.A.	Asia e University, Malaysia	Aug 2025	GPA: 3.65/4.00
B.E. Computer Engineering	Tribhuvan University, Nepal	Sep 2022	GPA: 3.64/4.00

CURRENT RESEARCH

Agentic AI for Cardiac Decision Support Aug 2026 – Present
Research Project | Advisor: Dr. I Hua Tsai | Dataset: MIT-BIH Arrhythmia (PhysioNet)

- Building an **agentic clinical decision-support system** for cardiac care, where specialized agents reason over calibrated model outputs and patient context; developed the ECG perception layer as a two-branch **1D CNN + RR-interval** arrhythmia classifier on **109K beats** using AAMI 5-class labeling.
- Ran **controlled ablations** under a strict inter-patient (DS1/DS2) protocol to isolate architecture from class-imbalance effects; evaluated **probability quality** (macro-F1, average precision, Brier score, ECE) for downstream agent reliability.

Fairness vs. Accuracy in Machine Learning Models May 2026 – Present
Research Project | Advisor: Dr. Gorkem Kar

- Built a **PyTorch** training method with a tunable accuracy–fairness trade-off and evaluated it across **5 real-world decision-making datasets** spanning criminal risk scoring, income, credit default, and health coverage, with **10 runs** per setting.
- Reduced demographic-group disparities by up to **93%** with **less than 0.5% accuracy loss**; identified a key trade-off where improving **group fairness** can worsen **individual fairness**.

PUBLICATIONS

- Nepal, T., Shrestha, Y., Ansari, K., Dadala, K., Tsai, I Hua, & Feng, A. Stress-Testing Biomedical Retrieval-Augmented Generation: A Diagnostic Evaluation under Noise, Conflicting Evidence, and Unanswerability.** *IEEE International Conference on E-health Networking, Application & Services (HealthCom), 2026*, New York, NY, Oct 19–21, 2026. [Accepted]
- Dadala, K., **Nepal, T.**, Tsai, I Hua. **CodeBERT at the Watering Hole for Dual-Layer URL and JavaScript Attack Detection.** *IEEE EIT 2026* (in proceedings). Presented: May 22, 2026.
- Nepal, T.** et al. **Generative Semantic Enhancement for Short-Video Feed Recommendation: A Converged Study on KuaiRec.** *IEEE International Conference on Big Data, 2026*, Phoenix, AZ, Dec 14–17, 2026. [Under Review]
- Nepal, T., Tsai, I Hua.** **What Hurts RAG in Biomedical QA: A Stress-Test Study of Noise, Conflict, and Unanswerability.** Lightning Talk & Poster, *Midwest Machine Learning Symposium (MMLS)*, Purdue University, Jun 25, 2026.

RESEARCH EXPERIENCE

Graduate Research Assistant Jan 2025 – Present
University of Central Missouri | Dr. I Hua Tsai | Lee's Summit, MO

- Designed and evaluated **RAG pipelines** for **BioASQ** biomedical QA; integrated **BM25**, dense embeddings, cross-encoder reranking, and evidence-constrained answer generation across factoid, list, and yes/no tasks.
- Developed **reproducible benchmarking** workflows and standardized evaluation protocols (**MRR, NDCG, EM, F1**); translated ablation results into **peer-reviewed** conference paper submissions.

RESEARCH PROJECT

Algorithmic Trading & Synthetic Index Modeling (Kalshi Markets) Aug 2025
Independent Research | Advisor: Dr. Gorkem Kar, UCM

- Built high-frequency trading system in **Rust** for short-term arbitrage in crypto prediction markets; trained **Deep Reinforcement Learning (DQN, PyTorch)** agents using a synthetic index model to improve trade timing and probability estimation.
- Performed **statistical backtesting** and risk-return analysis; deployed fault-tolerant system with real-time **Streamlit** analytics dashboard for multi-factor strategy evaluation.

TECHNICAL SKILLS

ML & NLP: RAG, LLMs (LLaMA-3, GPT, Claude), **HuggingFace Transformers, PyTorch**, TensorFlow, Scikit-learn, Prompt Engineering, Fine-tuning, **DRL (DQN)**, Multi-hop QA, Knowledge Graphs

Information Retrieval: BM25, Dense Embeddings, **Cross-Encoder Reranking**, MRR, NDCG, Recall@K, Annotation & Data Curation

Programming: Python, R, **SQL, Rust**, Git

Cloud & MLOps: **AWS** (SageMaker, Lambda, S3, Redshift, CloudFormation), Docker, **FastAPI**, LangChain, **Streamlit**, Pydantic

INDUSTRY EXPERIENCE

Financial Data Analyst Dec 2022 – Nov 2024
Genese Solution Pvt. Ltd., Nepal

- Built LLM-driven automation tool using **Anthropic API + AWS CloudFormation** to deploy infrastructure from natural language prompts; improved provisioning efficiency by **20%**.
- Developed **credit risk forecasting** models (Scikit-learn, TensorFlow) improving predictive accuracy by **15%**; applied **HuggingFace** transformer NLP for financial text analysis, reducing deployment defects by **90%**.

AWARDS & GRANTS

UCM Graduate Student Travel Fund, awarded twice (2026) for conference travel to IEEE EIT 2026 and IEEE HealthCom 2026

Merit-Based Semester Scholarships for Academic Excellence, Lalitpur Engineering College, Tribhuvan University, 2021 & 2022

Data Insight's Data Scientist Program Scholar, British Columbia, Canada, Dec 2021 – Dec 2022

CERTIFICATIONS

AWS Certified AI Practitioner, Amazon Web Services, Verify Credential, Valid through Nov 2028

General Registered Engineer, Nepal Engineering Council (Act 1999 AD), Registration No. 85011, Jul 2024 – Present